FIT3162/3164 Final Project Presentation and Software Demo

Project Title: Intelligent Network Intrusion Detection System

Presented by:

Group MCS13

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Overview

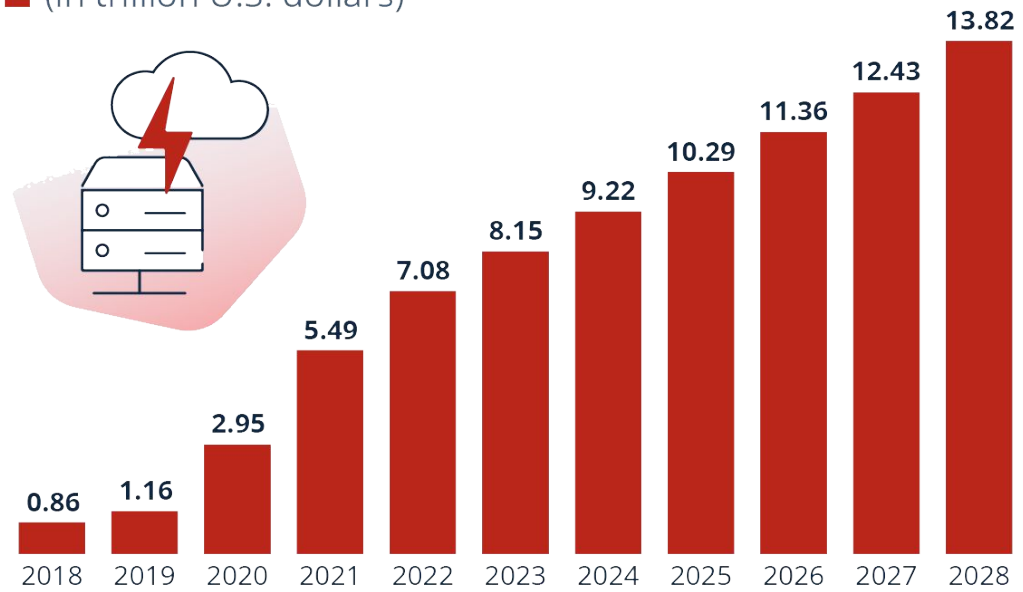
- *Introduction*
  - *AI Model Training*
- *Project Details*
  - *Scope*
- *Project Demonstration*
- *Conclusion*

Introduction

Cybersecurity today and in the future

Cybercrime Expected To Skyrocket

Estimated annual cost of cybercrime worldwide (in trillion U.S. dollars)



Problem statement

Traditional Intrusion Detection Systems cannot keep up with modern cyber attacks.

Opportunity Statement

Training a pre-existing model and improving performance.

Relevant research

No.	Author	Research Category	Research summary
1	Sarker et al. 2020	Development of new machine learning models for modern Intrusion detection.	Optimization for efficient network monitoring.
2	Abu Al-Haija & Zein-Sabatto, 2020		Deep-learning-based detection and classification system.
3	Ren et al., 2023		Optimization for data-preprocessing.
4	Vadivelan et al., 2022	Comparison and reviews of pre-existing models and strategies for intrusion detection.	Breakdown of major cyber attack categories.
5	Zehra et al., 2023		Accuracy testing for deep learning techniques.
6	Talaei Khoei & Kaabouch, 2023		Optimizing current ML models.
7	Liu et al., 2024		Comparing supervised, unsupervised, and hybrid models.
8	Badr, 2022	Hybrid research on both developing and comparing AI models for intrusion detection.	Suggests Q-learning as a reinforcement learning method.
9	Ullah et al., 2025		Evidence for Deep Neural Network performance.
10	Rai et al., n.d.		Network based intrusion detection.
11	Parasuraman et al., n.d.		naive Bayes and self organizing maps.
12	Salem et al., 2024		Suggest DL and ML hybrid solution

Research Commonalities

1. Data pre-processing optimization

- Optimizing feature extraction method.

2. Accuracy testing of model

- Comparing the anomaly detection accuracy with other methods.

3. Suggesting different algorithms

- Comparing performance of different algorithms used for intrusion detection

AI Model training

- an unsupervised machine learning – kitNET
- Basic dataset – CIC-IDS 2017

ML Training method: Cross validation

Train model on a dataset of 200,000 packets.

Tested on 4 sets of 50,000 packets.

Project Details

Main Goals

Software should:

- monitor network traffic
- detects suspicious activity
- machine learning

- Integrate unsupervised model into our IDS
- Tested on CIC-IDS2017 dataset

Providing an accessible interface for users.

Project Details

Limitations

1. Future threats cover many possible scenarios, which is difficult to account for

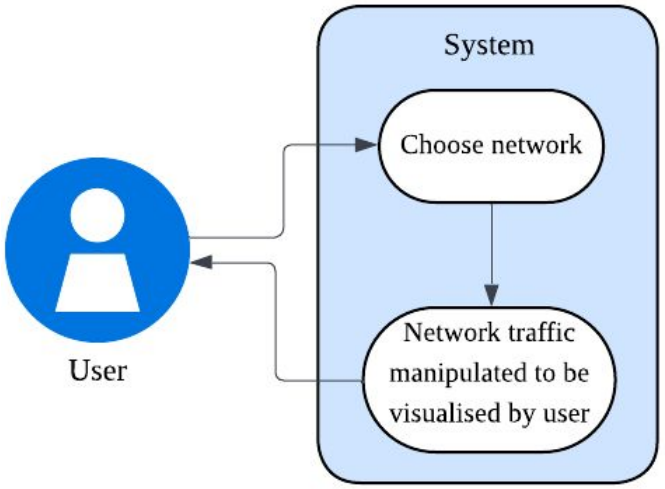
2. Long training times due to large datasets

3. Difficulties in reaching desired performance metrics

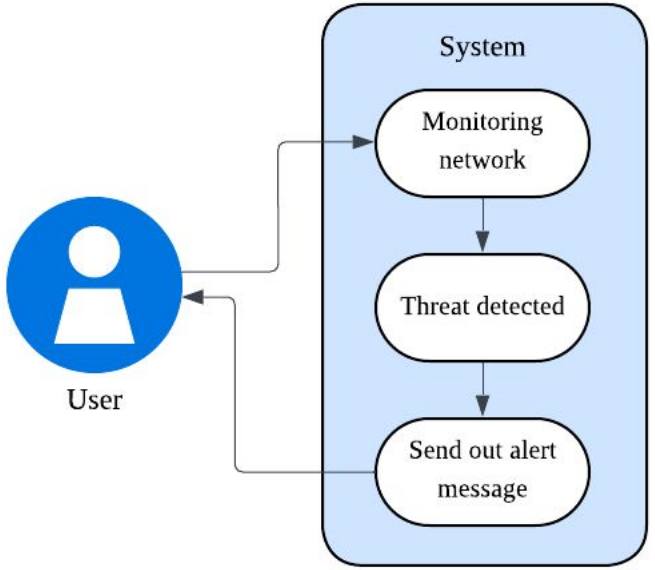
Project Details: Scope

Use Cases

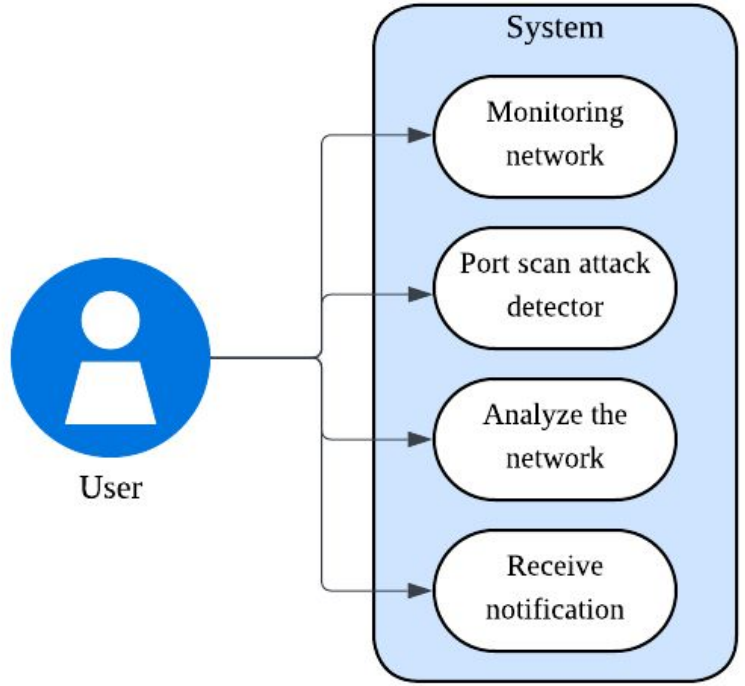
Choosing a network to monitor.



Visualising network traffic on a device.



Detecting bad traffic mixed and pushing out alert messages.



Project Details: Scope

Test Approaches

Unit Testing:

a. Testing each feature

b. Testing happy paths and error conditions

c. Testing real cases and edge cases

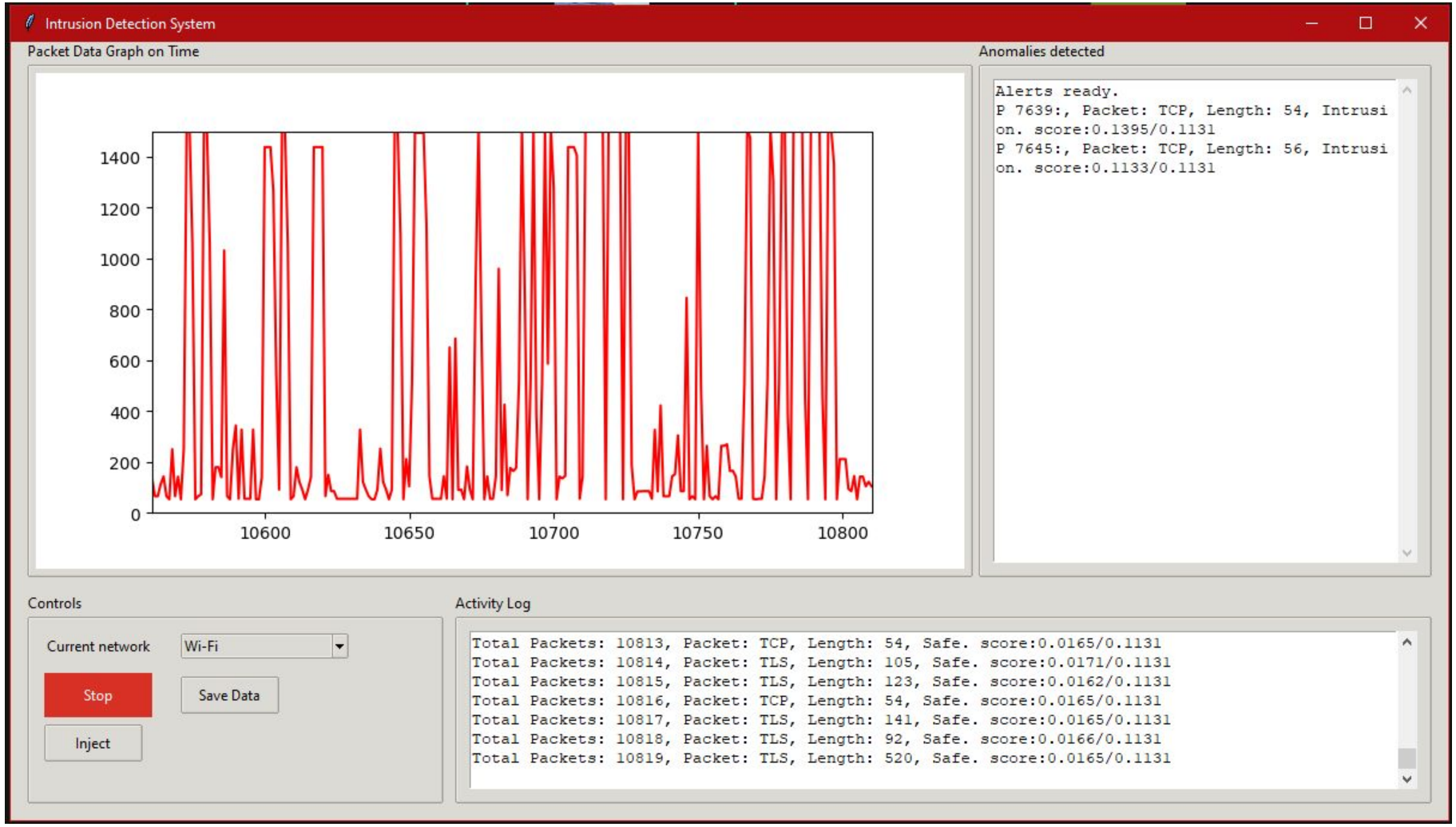
Application:

a. Test integration of features working together

Project Demonstration

Jeremy Koh Chi Minn 13.

Project Demonstration



Conclusion

Unachieved goals

Adaptation requirements not met

- R7: Software does not output performance metrics
- R8: Admin cannot flag false positives
- R9: Does not improve upon existing machine learning model

Integration of supervised model would require too many steps

- Poor performance
- Not ideal for real-time intrusion detection

Conclusion

Achieved goals

All other requirements have been met

Functionality

Performance

Usability

Conclusion

Future work proposal

Machine learning model that can train in real time and adapt to local network traffic data

Conclusion

Key takeaway (effective teamwork)

Set realistic goals for the team as a whole

- Not everyone has the same skill set or knowledge

Respect for differing skills and perspectives

- Make up for each other's weaknesses
- Delegate tasks according to individual strengths

Communication is key

- Keep each other in the loop
- Maintain a positive attitude

Thank you

Supervisor Sign-off



Muhammad Fermi Pasha